# LABSHEET 4

## A. Arithmetic Operators

### 1. Perform addition of two numbers using +.

```
SELECT 15 + 25 AS Addition_Result;
```

### 2. Perform subtraction using -.

```
SELECT 50 - 18 AS Subtraction_Result;
```

### 3. Perform multiplication using *.

```
SELECT 12 * 8 AS Multiplication_Result;
```

### 4. Perform division using /.

```
SELECT 45 / 5 AS Division_Result;
```

### 5. Calculate the total amount as Price × Quantity.

```
SELECT Product_ID, Product_Name, Price, Quantity,
 Price * Quantity AS Total_Amount
FROM Product;
```

## B. Logical Operators

### 6. Demonstrate AND with two conditions.

```
SELECT * FROM Student
WHERE Age >= 18 AND City = 'Delhi';
```

### 7. Demonstrate OR with two conditions.

```
SELECT * FROM Student
WHERE City = 'Delhi' OR City = 'Meerut';
```

### 8. Demonstrate NOT with a condition.

```
SELECT * FROM Student
WHERE NOT City = 'Delhi';
```

### 9. Find students belonging to a specific course AND city.

```
SELECT * FROM Student
WHERE Course = 'BCA' AND City = 'Delhi';
```

### 10. Find employees belonging to either of two departments.

```
SELECT * FROM Employee
WHERE Department = 'IT' OR Department = 'HR';
```

## C. Comparison Operators

### 11. Demonstrate the = operator.

```
SELECT * FROM Student
WHERE Course = 'BCA';
```

### 12. Demonstrate the > operator.

```
SELECT * FROM Employee
WHERE Salary > 40000;
```

### 13. Demonstrate the < operator.

```
SELECT * FROM Employee
WHERE Salary < 30000;
```

### 14. Demonstrate >= and <=.

```
SELECT * FROM Employee
WHERE Salary >= 30000 AND Salary <= 50000;
```

### 15. Demonstrate the <> / != operator.

```
SELECT * FROM Student
WHERE City <> 'Delhi';

-- MySQL also supports !=
SELECT * FROM Student
WHERE City != 'Delhi';
```

## D. Special Operators

### 16. Demonstrate BETWEEN for salary range.

```
SELECT * FROM Employee
WHERE Salary BETWEEN 30000 AND 50000;
```

### 17. Demonstrate IN for selecting multiple cities.

```
SELECT * FROM Student
WHERE City IN ('Delhi', 'Meerut', 'Roorkee');
```

### 18. Demonstrate NOT IN.

```
SELECT * FROM Student
WHERE City NOT IN ('Delhi', 'Meerut');
```

### 19. Demonstrate LIKE for names beginning with a particular letter.

```
SELECT * FROM Student
WHERE Name LIKE 'A%';
```

### 20. Demonstrate IS NULL and IS NOT NULL.

```
SELECT * FROM Student
WHERE City IS NULL;
```

```
SELECT * FROM Student
WHERE City IS NOT NULL;
```

## E. Set Operations

### 21. Demonstrate UNION using two compatible queries.

```
SELECT City AS Place FROM Student
UNION
SELECT Location AS Place FROM Department;
```

### 22. Demonstrate UNION ALL.

```
SELECT City AS Place FROM Student
UNION ALL
SELECT Location AS Place FROM Department;
```

### 23. Demonstrate INTERSECT where supported by the DBMS.

```
SELECT DISTINCT s.City
FROM Student AS s
WHERE s.City IS NOT NULL
 AND EXISTS (
 SELECT 1 FROM Department AS d
 WHERE d.Location = s.City
 );
```

## 24. Demonstrate EXCEPT/MINUS where supported.

```
FROM Student AS s
WHERE s.City IS NOT NULL
 AND NOT EXISTS (
 SELECT 1 FROM Department AS d
 WHERE d.Location = s.City
 );
```

## 25. Use arithmetic, logical, comparison and special operators together.

```
SELECT Emp_ID, Emp_Name, Salary,
 Salary * 1.10 AS Salary_After_10_Percent_Raise
FROM Employee
WHERE (Department = 'IT' OR Department = 'HR')
 AND Salary BETWEEN 30000 AND 50000
 AND Salary >= 35000;
```